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① Conversion

a) $\underset{7\,6\,5\,4\,3\,2\,1\,0}{11110010}$

$2^7 \cdot 1 + 2^6 \cdot 1 + 2^5 \cdot 1 + 2^4 \cdot 1 + 2^3 \cdot 0 + 2^2 \cdot 0 + 2^1 \cdot 1 + 2^0 \cdot 0 =$

$= 128 + 64 + 32 + 16 + 2 = 242$

b) $\underset{7\,6\,5\,4\,3\,2\,1\,0}{11111111}$

$2^7 \cdot 1 + 2^6 \cdot 1 + 2^5 \cdot 1 + 2^4 \cdot 1 + 2^3 \cdot 1 + 2^2 \cdot 1 + 2^1 \cdot 1 +$

$+ 2^0 \cdot 1 = 128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 =$

$= 255$

c) $\underset{7\,6\,5\,4\,3\,2\,1\,0}{11000000}$

$2^6 + 2^5 = 64 + 32 = 96$

~~c) $\underset{7\,6\,5\,4\,3\,2\,1\,0}{11000000}$~~

~~$2^7 \cdot 1 + 2^6 \cdot 1 = 128 + 64 = 192$~~

②

a) $\underset{6\,5\,4\,3\,2\,1\,0\;\;-1\,-2\,-3\,-4}{1011010,1010}$

$2^6 \cdot 1 + 2^4 \cdot 1 + 2^3 \cdot 1 + 2^1 \cdot 1 + \frac{1}{2} + \frac{1}{8} =$

$= 64 + 16 + 8 + 2 + 0,5 + 0,125 =$

$= 90,625$

b) $\underset{6\,5\,4\,3\,2\,1\,0\;\;-1\,-2\,-3\,-4\,-5}{1011100,10101}$

$2^6 + 2^4 + 2^3 + 2^2 + 0,5 + 0,125 + 0,03125 =$

$= 64 + 16 + 8 + 4 + 0,65625 = 92,65625$

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c) $\underset{6\,5\,4\,3\,2\,1\,0}{1\,1\,1\,1\,1\,1}.\underset{-1\,-2\,-3\,-4\,-5}{1\,1\,1\,1\,1}$

$2^6+2^5+2^4+2^3+2^2+2^1+1+0,5+0,25+0,125+$
$+0,0625+0,03125=$
$=64+32+16+8+4+2+1+0,96875=$
$=126,96875$

(3)

a) 49 | 1
24 | 0
12 | 0
6 | 0
3 | 1
1 | 1
0 |

$49 = 110001$ (in binary)

b) 60 | 0
30 | 0
15 | 1
7 | 1
3 | 1
1 | 1
0 |

$60 = 111100$ (in binary)

(4)

d) $0,456 < 0.5 \rightarrow 0$
$0.456 - 0.25 = 0.206 \rightarrow 1$
$0.206 - 0.125 = 0.081 \rightarrow 1$
$0.081 - 0.0625 = 0.0185 \rightarrow 1$
$0.0185 < 0.015625 \rightarrow 0$

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$0.0185 - 0.015625 = 0.002875 \rightarrow 1$

$0.002875 < 0.0078125 \rightarrow 0$

$0,456_{10} \simeq 0,01110_2$ (approximate value)

b) $0.8732 - 0.5 = 0.3732 \rightarrow 1$

$0.3732 - 0.25 = 0.1232 \rightarrow 1$

$0.1232 < 0.125 \rightarrow 0$

$0.1232 - 0.0625 = 0.0607 \rightarrow 1$

$0.0607 - 0.03125 = 0.2945 \rightarrow 1$

$0.02945 - 0.015625 = 0.013825 \rightarrow 1$

$0.013825 - 0.0078125 = 0.0060125 \rightarrow 1$

$0.8732_{10} \simeq 0.111101_2$ (approximate value)

(1) Binary Arithmetic.

a) $100 + 11 = 111$

$$\begin{array}{r} 100 \\ +\ 11 \\ \hline 111 \end{array}$$

b)
$$\begin{array}{r} {\scriptstyle 1\ 1\ 1} \\ 111 \\ +\ 101 \\ \hline 1100 \end{array}$$

c)
$$\begin{array}{r} {\scriptstyle 1\ 1\ 1\ 1} \\ 1111 \\ +\ 1111 \\ \hline 11110 \end{array}$$

(2)

a)
$$\begin{array}{r} 1111 \\ -\ 11 \\ \hline 1100 \end{array}$$

b)
$$\begin{array}{r} 1101 \\ -\ 101 \\ \hline 1000 \end{array}$$

c)
$$\begin{array}{r} {\scriptstyle 1}10000 \\ -\ 1111 \\ \hline 10001 \end{array}$$

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③

a) 1100 × 101 = 1100 + 0000 + 1100 = 111100

b) 1110 × 1110 = 10000 + 101110 + 11110 + 1110 = 11000100

c) 1111 × 1100 = 10000 + 10000 + 1111 + 1111 = 10110100

④

a) 1010 ÷ 10 = 101

1010	10
−10	101
010	
−10	
0	

b) 1111 ÷ 101 = 11

1111	101
−101	11
101	
−101	
0	

⑤ Signed Numbers

a) −34

34	0
17	1
8	0
4	0
2	0
1	1
0	

$100010_2 = 34_{10}$

−34 ⇒ 10100110

1's complement:

11011101

b) −99

99	1
49	1
24	0
12	0
6	0
3	1
1	1
0	

99 = 1100011

1's complement:

10011100

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(2)

a) -68

68	0
34	0
17	1
8	0
4	0
2	0
1	1

$+1000100$

1's complement:

10111011

2's complement:

$$\begin{array}{r} 10111011 \\ +\quad\quad\quad 1 \\ \hline 10111100 \end{array}$$

Answer: 10111100

b) -125

125	1
62	0
31	1
15	1
7	1
3	1
1	1
0	

01111101

1's complement:

10000010

2's complement:

$$\begin{array}{r} 10000010 \\ +\quad\quad\quad 1 \\ \hline 10000011 \end{array}$$

Answer: 10000011